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TITLE

Inquiry into variation in clinical presentation in SCD patients with different hemoglobin variants along with genetic-analysis. Qualitative study in district of union-territory in western India.

INTRODUCTION

Sickle cell disease is an autosomal recessive blood disorder marked by the formation of abnormal hemoglobin molecules that leads to crescent or sickle shaped red blood cells. It affects millions of people all over the globe, specifically those of African, Mediterranean, Middle Eastern, and South Asian descent [1].

The main cause of SCD is a mutation in the beta-globin (HBB)gene, which helps in production of hemoglobin [2]. The altered shape of red blood cells contributes to various health complications associated with SCD.

In India HbS is markedly seen in schedules caste, scheduled tribe, and other backward population in western, central and parts of eastern and southern India. The prevalence of HbS is 2-40%.[4].

Hemoglobin D^{Punjab} (HbD^{Punjab}) (also known as $HbD^{Los\text{-}Angeles}$, $HbD^{Portugal}$, $HbD^{North\ Carolina}$, $D^{Oak\ Ridge}$ and $D^{Chicago}$) is a hemoglobin variant due to point mutation in codon 121 (GAG→GAC) of β globin chain resulting in substitution of amino acid glutamic acid to glycine.[5]. It is a widely distributed hemoglobin with a relatively low prevalence of 0.86% in the Indo-Pak subcontinent and 1–3% in north-western India. [6], [7], [8], [9], [10], [11], [12].

Inheritance of Hb D-Punjab and HbS together can cause vaso-occlusive events including acute chest and stroke, which is similar to homozygous HbS inheritance that manifests clinically as sickle cell disease [13],[14].

The prevalence of hemoglobin E (HbE) in India ranges from 3% to 4% in West Bengal in the east and 4% to 51% in the north eastern region. It is most common abnormal variant in Southeast Asia. The HbE mutation is (β^{26} GAG → AAG), substitution of glutamic acid by lysine. An alternative splice site is created with insufficient synthesis of β^{E} chain. Therefore, it's phenotype is similar to mild form of β thalassemia [15].

Hemoglobin C is due to point mutation (β^{6} GAG → AAG), substitution of glutamic acid by lysine.

Alpha thalassemia and Beta thalassemia is caused by reduced or absent synthesis of alpha and beta globin chains respectively.

Inheritance of these with HbS results in variable clinical manifestations.These structural variants are seen across various regions of India, and their interaction is seen due to migration of people for many reasons like better livelihood. There are very few studies and reports on interaction of these commonly seen Hb variants and the phenotypic–genotypic presentation of these cases is important for genetic counseling and management.

OBJECTIVES

- Primary Objective

1) To investigate the influence of various hemoglobin variants, including HbS, HbD, HbE, alpha-thal, beta-thal and their combinations with HbS, on the clinical presentation of SCD.

2) To investigate the influence of various hemoglobin variants, including HbS, HbD, HbE, alpha-thal, beta-thal and their combinations with HBS, on the severity of SCD.

- Secondary Objective

1) Perform genetic analysis to identify mutations and genetic markers associated with different hemoglobin variants and their combination with HbS.
2) Establish correlations between specific genetic mutations and the clinical severity of hemoglobinopathies involving HbS.

METHODOLOGY

Study area-This study will be done in a district of union territory of western India with a majority of tribal and migrant population.

Study design- The present study is qualitative study.

Study population-The study population will be confirmed cases of sickle cell disease in our study area.

Sample Size - Following groups will be included in this study

1. All the subjects who are confirmed cases of sickle cell disease with double compound heterozygous inheritance of other hemoglobin variants (HbD, HbE, HbC, Alpha-Thal, Beta-Thal) with HbS. Atleast 20 cases.

2. The subjects who are confirmed cases of sickle cell disease in homozygous state. Approximately 20 cases.

Total around 40 cases till saturation is obtained.

Sampling technique –

All the patients with hemoglobin variants will be identified from the list of total sickle cell disease patients for the study area. All the patients with hemoglobin variants will be included in the study. Patient with sickle cell homozygous state will also be included in the study.

Study duration: From 01st August 2024 to 01st August 2025. (3 months in the said timeline)

Study tool: Independent variables will be studied separately by suitable questionnaires.

Independent variables:

Genetic Analysis: Done by taking blood sample and performing PCR (Polymerase Chain Reaction).

Patients will be enquired regarding their:

Age- The age of the subject will be collected through the questionnaire, interview, medical records and other documents having date of birth. Age will be written in completed years.

Gender-The gender of the subject will be collected through the questionnaire, interview, medical records and other documents specifying gender.

Educational qualification -The subject's education status will be collected by the questionnaire and personal interview.

Socio economic status- The socio-economic status of the subject will be assessed using an updated modified BG Prasad classifications

Residential area-the residential area of the subject will be determined by the questionnaire, personal interview or any other document in which the address of the subject is given. It will be determined in the form of urban or rural areas.

Number of family members - It will be collected by the questionnaire, personal interview or other documents specifying the number of people in the family.

Caste-The caste of the subject is determined by the questionnaire, personal interview or other documents specifying the number of people in the family

Religion-The data about the subject's religion will be collected by the questionnaire and personal interview.

Type of fuel used for cooking- This data will be collected by the questionnaire and personal interview.

Employment status- The employment status of the subject will be collected from the questionnaire, personal interview or other documents specifying the number of people in the family. It will be collected as employed and unemployed.

Occupation- occupation will be determined by the questionnaire and personal interview.

Any addiction – Alcohol, Smoking, tobacco or others.

Awareness about sickle cell disease? – Whether the patient is aware of having the disease or not.

Age at which diagnosed for SCD? – At which age the patient was diagnosed for having SCD.

Family screening- Whether parents, siblings, children and other family members are screened for SCD.

Treatment- Whether the patient is undergoing any treatment. Whether the patients take medicines, supplements or any other treatment.

Local Remedy- Any other local remedy taken?

Complication- Any complications due to treatment will be assessed.

Hospital admission- Major surgeries, pain crisis, blood transfusions and any other complaints.

Lifestyle & Diet changes- changes in lifestyle and diet will be assessed.

Impact on quality of life- It will be assessed by PQLI.

Data Collection

Data collection will involve multiple sources, ensuring comprehensive and reliable information:
1) Patient Demographics: Age, sex, race, ethnicity, and socioeconomic status will be recorded from medical records.

2)Medical History: Past medical history, including previous SCD-related complications, hospitalizations, and treatments, will be collected through review of medical records.

3) Clinical Data: Hematological parameters (e.g., hemoglobin levels, red blood cell count), organ function (e.g., liver, kidney, spleen), and symptom severity (e.g., pain scores, frequency of vaso-occlusive crises) will be obtained from medical records and clinical assessments.

4) Treatment History: Information on past and current SCD treatment modalities, such as hydroxyurea, blood transfusions, and gene therapy, will be collected from medical records.
5)Genetic Data: DNA samples will be collected to determine the hemoglobin genotype of each participant, using advanced genetic testing methods.

6)Patient Perspectives: Questionnaires and interviews will gather subjective data on patient experiences, quality of life, and socioeconomic factors influencing their health outcomes.

Data Analysis: In depth Interviews will be recorded. It will be then transcripted and translated. Themes will be identified and thematic analysis will be done.

Inclusion criteria: All the patients with homozygous or double compound heterozygous disease states will be included in the study.

Exclusion criteria: Patients without a written consent will be excluded from the study

Ethical consideration: Ethical approval will be taken from the ethics committee. The consent will be taken from concerned authorities. The identity of the respondents will be kept anonymous. Respondents will be properly counselled about their issues.

IMPLICATION

The findings of this study have the potential to significantly impact the clinical management of SCD.

Implications include:

- Improved diagnosis and risk stratification of SCD patients based on their hemoglobin genotypes.
- Development of personalized treatment plans tailored to specific hemoglobin variants.
- Enhanced understanding of the socioeconomic factors influencing SCD health outcomes, leading to targeted interventions.
- Advancement of research in gene therapy and other innovative therapeutic approaches for SCD.

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